

### Sound detector

A cochlear implant has many components. The external section captures sound waves and produces electrical signals, which the brain can interpret. These signals pass into the internal section of the implant, which transmits them to the cochlea, then to a nerve that carries them straight to the brain.

The external transmitter is kept in place with a magnet and passes signals to the receiver inside the body.

A microphone and speech processor sit above the ear. The microphone produces signals from sound it picks up, and the processor converts them to signals the brain will understand.

A receiver sits inside the body and picks up the signals.

A tiny electrode array runs from the receiver to the cochlea and passes along the signals.

The auditory nerve passes electrical information to the brain.

The cochlea

The ear canal is where sounds usually enter, but a cochlear implant bypasses this.

**There are more than 250,000 people with cochlear implants worldwide.**

